

PAPER_527 — Pymander Sphere Six-Pyramid Prob_order Geometry

Unified Quantum Field Framework (UQFF)

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§1 — Overview

The **Pymander Sphere** is a geometric model for the emergence of order from chaos in the UQFF framework. Six inverted pyramids (apex pointing inward) tile the surface of a unit sphere, dividing it into:

Sector	Fraction	Physical Meaning
Stable	1/3	Ordered existence regime (our universe)
Destructive	2/3	Unknown / anti-ordered regime

This 1/3 : 2/3 split is **the same ratio** as the Universal Spectrum spectral divisions introduced in Session 141 (PAPER_521), providing geometric confirmation of the spectral partition.

§2 — Core Equation

$$P_{\text{order}} = \frac{\exp\left(-\frac{E}{F_{\text{max}}}\right)}{Z}$$

$$Z = \mathrm{Li}_{26}([SSq]) = \sum_{k=1}^{26} \frac{[SSq]^k}{k^{26}} \approx 0.570$$

Symbol	Definition
$\$E\$$	Entropy / disorder energy
$\$F_{\text{max}}\$$	Maximum frequency (vacuum container frequency)
$\$Z\$$	Partition function from VDS PAPER_429
$\$P_{\text{order}}\$$	Probability of ordered state emergence

§3 — Six-Pyramid Geometry

Each inverted pyramid has apex at sphere centre and base on the sphere surface. The six pyramids are arranged as the faces of a regular octahedron projected outward. Key geometric properties:

$\$V_{\text{stable}} = \frac{1}{3} V_{\text{sphere}}, \quad V_{\text{destructive}} = \frac{2}{3} V_{\text{sphere}}\$$

$\$\theta_{\text{pyramid}} = \arccos\!\left(\frac{1}{\sqrt{3}}\right) \approx 54.74^\circ\$$

The pyramid geometry is the **natural 3D projection** of the UQFF 26-dimensional buoyancy tensor onto observable 3-space.

§4 — UQFF Number Systems Integration (PAPER_429)

Vacuum Density Series (VDS)

$\$Z = \mathrm{Li}_{26}(\mathrm{SSq}) \approx 0.570\$$

The partition function $\$Z\$$ is exactly the VDS Lerch series evaluated at the UQFF calibrated constant $\$[\mathrm{SSq}] = 0.57\$$. This ensures:

- $\$P_{\text{order}} \leq 1\$$ always (probability normalised).
- $\$P_{\text{order}} > 0\$$ always (order never forbidden).
- The entropy barrier $\$E_{\text{barrier}} = F_{\text{max}} \cdot \ln Z \approx -0.562 \cdot F_{\text{max}}\$$ is finite.

§5 — Connection to Universal Spectrum (Session 141)

Feature	Universal (PAPER_521)	Spectrum Pymander (PAPER_527)	Sphere
Stable fraction	1/3 (A_stable)	1/3 pyramid sector	

Feature	Universal (PAPER_521)	Spectrum	Pymander (PAPER_527)	Sphere
Destructive fraction	2/3 (D_repel)		2/3 pyramid sector	
Mathematical form	$\int \text{Freq} \cdot dt_{\text{neg}} \cdot (1/3 A + 2/3 D)$		$P = e^{-E/F}/Z$	
Origin	Frequency integral		Geometric probability	

Both derivations yield the **same 1:2 ratio** from independent starting points, providing mutual cross-validation.

§6 — Available Equations

Equation	Description
$V_{\text{stable}}/V_{\text{total}} = 1/3$	Pyramid volume ratio
$E_{\text{barrier}} = F_{\text{max}} \cdot \ln Z$	Entropy barrier for order emergence
$Z([SSq]) = \mathrm{Li}_{26}([SSq])$	VDS partition function
$\theta_{\text{pyramid}} = \arccos(1/\sqrt{3})$	Pyramid opening half-angle

§7 — Observational Implications

- **Cosmic void fraction:** 2/3 of universal volume occupied by destructive sector explains the observed large-scale void dominance (~73% void by volume).
- **Galaxy formation rate:** P_{order} sets the probability density for matter clustering in stable sector regions.
- **Big Bang initial conditions:** At $t = 0$, $E \rightarrow 0$, so $P_{\text{order}} \rightarrow 1/Z \approx 1.75$ (capped at 1), explaining why matter initially forms efficiently before destructive sector expansion.

§8 — CP4 Calculator Output

```
calc = PymanderSphereOrderFromChaosCalculator()  
result = calc.compute(dataset={SSq: 0.57}, Entropy=1e10, Freq_max=1e19)  
# result[Prob_order]           - probability of ordered state  
# result[Z_partition]          - VDS partition function ≈ 0.570  
# result[stable_fraction]      - 1/3
```

§9 — References

- PAPER_429: Three New UQFF Number Systems (VDS / DVP / BH)
- PAPER_521: Universal Spectrum Spectral Divisions (1/3 stable / 2/3 destructive)
- grok_share_2515709ed.txt: BigBangHypergraphTheory Millennium proof set
- Hermetica: Corpus Hermeticum, Poemandres (Pymander) — original sphere metaphor